

HEALTHCARE AI RESEARCH GROUP

Predictive Cardiology

Applying Machine Learning Algorithms to Foresee and Prevent Cardiovascular Events

Global Burden of Heart Disease

| The Scope of the Crisis

Cardiovascular Diseases (CVDs) remain the leading cause of death globally, accounting for over 17.9 million annual deaths.

Up to 80% of premature heart attacks and strokes are preventable through early detection, lifestyle alteration, and clinical intervention.

| Early Diagnostic Hurdles

Traditional diagnostic procedures are often costly, reactive, and require intensive manual laboratory screenings.

AI-driven prediction pipelines leverage existing electronic health records to proactively flag asymptomatic high-risk patients.

Transitioning to Proactive Care

Early Risk Identification

Machine Learning models shift cardiology from emergency response to predictive prevention.

- ✓ **Continuous Analysis:** Sifts through high-dimensional clinical parameters in real-time.
- ✓ **Non-Obvious Patterns:** Uncovers multivariate statistical dependencies that escape standard physician rubrics.
- ✓ **Scalability:** Instantly processes thousands of complex patient records at negligible incremental cost.

600 × 400

Standard Research Repositories

Dataset Registry	Standard Dimensions	Core Attributes	Primary Research Utilization
Cleveland Database (UCI)	303 Patient Records	14 Key Features	Gold-standard for binary classification benchmark
Framingham Study	5,209 Long-Term Records	Demographics, Lifestyle	Predicting 10-year risk of cardiovascular disease
Statlog Heart Dataset	270 Clinical Observations	13 Features (Structured)	Evaluating strict cost-sensitive classification models
Hungarian Dataset	294 Medical Records	Laboratory Markers	Validating classifiers on sparse, missing data cohorts

Key Baseline Features (Part I)



Age & Gender

Biological age and physiological gender profile represent standard, crucial inputs for foundational stratification within cardiac hazard models.



Resting BP

Systolic blood pressure measured in mm Hg on clinical admission provides critical indications of hypertension and arterial wall stress.



Cholesterol

Serum cholesterol levels in mg/dl measure circulating lipids, directly aiding predictions of plaque accumulation and risk of atherosclerosis.

Clinical Diagnostic Indicators



Resting ECG Results

Reveals abnormal conduction, left ventricular hypertrophy, or ST-T wave anomalies indicating structural strain.



Oldpeak Metric

Measures ST-segment depression induced by physical stress relative to rest, signifying coronary narrowing.



Exercise-Induced Angina

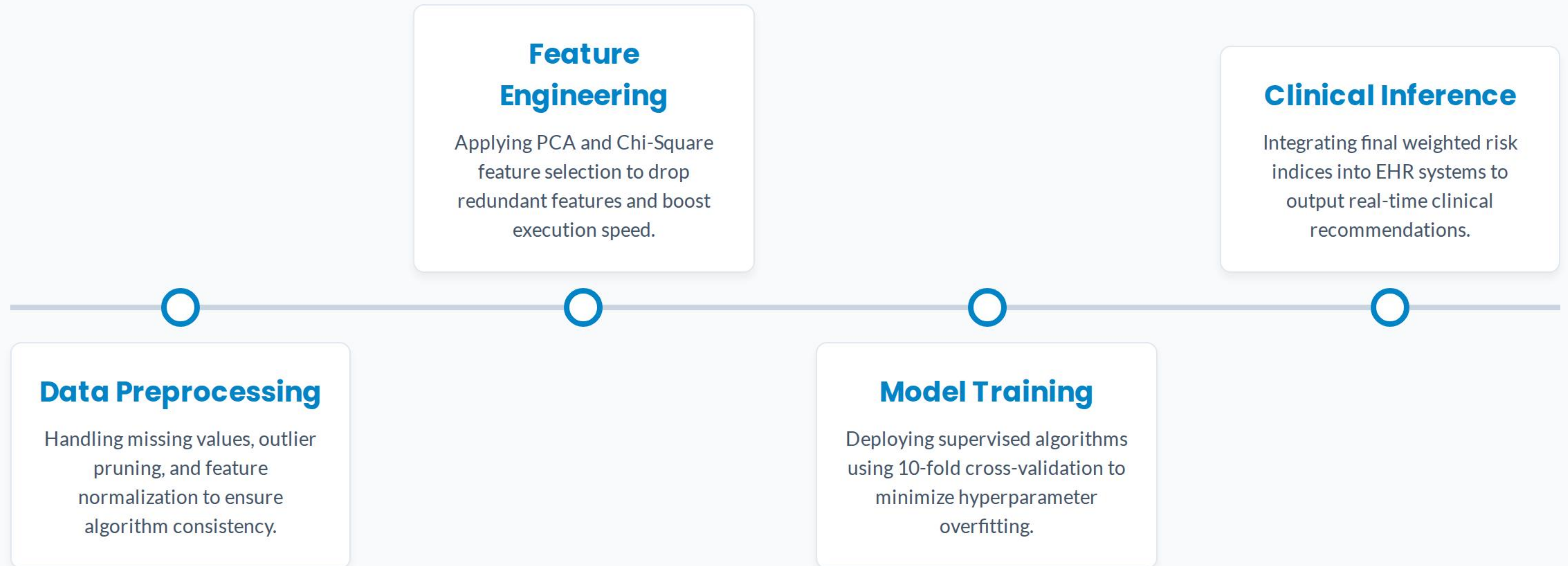
Identifies if physical exertion induces heart chest discomfort (angina), indicating ischemic limitations.



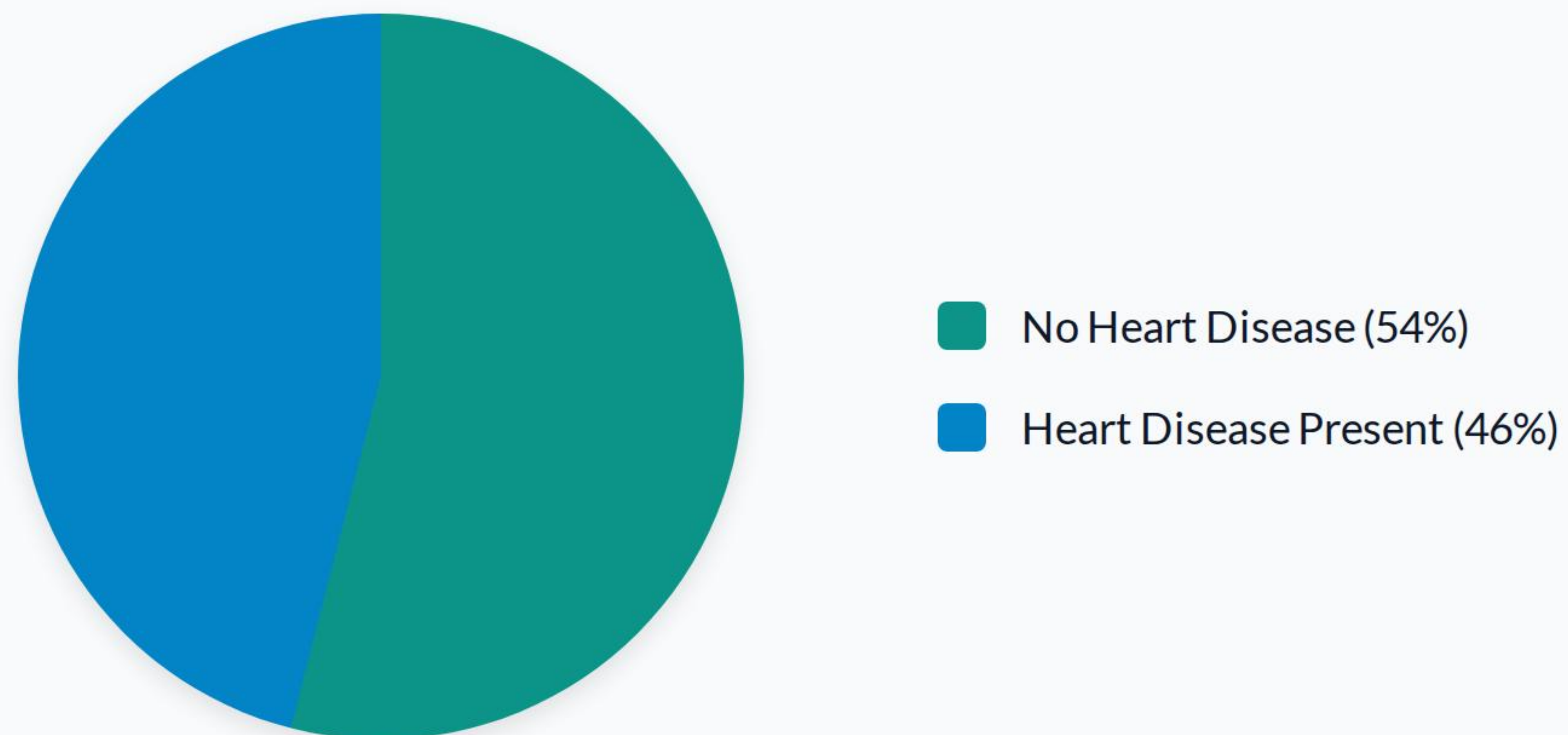
ST-Segment Slope

The orientation of the peak exercise ST-segment slope (upsloping, flat, downsloping) serves as a marker for ischemia.

Predictive System Architecture



Cleveland Dataset Risk Balance



The standard Cleveland cohort exhibits a highly balanced binary distribution, providing supervised learning models with optimal ground truth representation for robust generalizability.

Core Classification Logic

| Logistic Regression

Calculates the probability of heart disease using a sigmoid function of linear inputs. Exceptional interpretability makes it a healthcare favorite.

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

| Support Vector Machine

Constructs an optimal separating hyperplane in high-dimensional space. Performs exceptionally well when dealing with non-linear clinical attributes using the RBF kernel trick.

The Random Forest Paradigm



600 × 400

Combining Decision Trees

Random Forest algorithms build multiple independent decision trees on randomly sampled bootstrapped datasets (Bagging).

By averaging the outputs of individual estimators through majority voting, the ensemble reduces variance, eliminates overfitting, and naturally handles missing laboratory data points without skewing clinical outputs.

Evaluation Metrics

91.8%

Maximum Classification Accuracy

Efficacy Benchmark Indicators

While accuracy represents model correctness across all classes, clinical applications prioritize alternative validation metrics:

- ✓ **Sensitivity (Recall):** Vital for minimizing false negatives, ensuring high-risk patients are not missed.
- ✓ **Precision:** Ensures clinical interventions are targeted, reducing artificial anxiety and resource fatigue.
- ✓ **AUC-ROC Curve:** Standard measure of classifier diagnostic discrimination power.

Performance Comparison



Based on benchmark evaluations of the Cleveland registry, ensemble-based Random Forests consistently outperform traditional single-classifier structures in overall diagnostic accuracy.

Real-Time EHR Integration

Connecting AI to Clinics

Deploying models into clinical workflows requires linking models directly into Electronic Health Records (EHR).

An API-driven pipeline extracts patient vital and diagnostic data automatically during clinical consultations, sending real-time danger alerts straight to primary care interfaces.

600 × 400

Future Research Frontiers

600 × 400

Continuous Streaming

Integrating real-time sensor streams from modern wearable devices to capture early transient arrhythmias.

600 × 400

Multi-Modal Analytics

Unifying clinical diagnostic metrics with patient genomics data to predict congenital predispositions.

600 × 400

Explainable AI (XAI)

Employing SHAP values to explain complex model recommendations, fostering medical practitioner trust.